

An Efficient CNN and RNN Hybrid Model for the Detection of Epileptic Seizures in EEG Signals

Zayneb Sadek

Digital Research Center of Sfax. Sfax
University
Sfax, Tunisia
zayneb.sadek123@gmail.com

Abir Hadriche

Digital Research Center of Sfax. Sfax
University
Regim Lab.ENIS. Sfax University
Sfax, Tunisia
Abir.hadriche.tn@ieee.org

Nawel Jmail

Digital Research Center of Sfax. Sfax
University
Miracle Lab. Sfax University
Sfax, Tunisia
naweljmail@yahoo.fr

Abstract— Electroencephalogram (EEG) is a widely used supplementary test in epilepsy diagnosis. While traditional methods and deep learning networks have shown considerable success, they often rely on extensive human effort, particularly in feature extraction. This study proposes a hybrid Convolutional Neural Network and Recurrent Neural Network (CNN-RNN) algorithm that can autonomously learn features from EEG data. The RNN component calculates dependency and continuity features from the CNN's intermediate layer output, which are then connected to the final fully connected network for classification prediction. The algorithm was trained and tested on four datasets from the University of Bonn. For binary classification tasks, the proposed CNN-RNN model achieved accuracies of 98.33% on dataset 1 and 100% on dataset 2. When tested on datasets 3 and 4, the model attained 98.00% and 100% accuracy, respectively. Notably, the algorithm demonstrated superior performance on datasets 2 and 4 compared to state-of-the-art methods. These results suggest that the proposed CNN-RNN model offers a promising approach for automated EEG analysis in epilepsy diagnosis, potentially reducing the need for manual feature extraction and improving classification accuracy.

Keywords—Epilepsy, EEG, CNN- RNN

I. INTRODUCTION

The brain is the source of all cognitive and sensory stimuli and controls the body's vital functions. The information processing function of its core unit is good. When it carries out these tasks, it generates complex and complicated biopotential signals [1]. Several techniques can be used to record these signals. These recordings, known as electroencephalograms (EEGs), are full of information about how the brain and other body organs function [1,2,3]. Additionally, thanks to this information, we can learn more about both sensory and cognitive stimulation [1].

EEG signals are used in this investigation due to their low cost and high information content. Furthermore, the abnormal activity of these signals is utilized to detect disease and provides valuable data for tracking those conditions. One of these diseases can be detected by EEG signals.

A disorder that EEG data can identify is epilepsy [4,5,6]. This disease is estimated to affect millions of people worldwide [7]. It also can be either controlled by medication or surgical removal of epileptic regions [8].

Epilepsy is a disease that manifests itself in the form of seizures. Epilepsy seizure is an abnormal electrical activity that occurs temporarily in nerve cells [3,9].

Examining the morphology of EEG signals from the clinical perspective helps neurologists identify the disease and estimate its extent. Reviewing long-term records to identify this disease using EEG data takes time and money. Therefore, automated seizure detection methods will greatly reduce the workload of neurologists monitoring extended EEG signals.

Numerous studies have leveraged the Bonn database maintained by the University of Bonn's Department of Epileptology for EEG signal analysis, employing diverse methodologies that yield varying results. For instance, Ullah et al. (2018) utilized an ensemble of Pyramidal One-Dimensional Convolutional Neural Network (P-1D-CNN) models, segmenting the 1024 dataset into smaller segments with a 50% overlap, achieving an impressive seizure detection accuracy of 99.1%[10]. In contrast, Hussein et al. (2018) approached the problem by converting EEG data into non-overlapping segments and employing a Long Short-Term Memory (LSTM) network, reporting a seizure detection accuracy ranging from 90.0% to 100%[11]. Similarly, Yuan et al. (2018) utilized wavelet transforms to create scalogram sequences, integrating features obtained through Global Principal Component Analysis (GPCA) and Stacked Denoising Autoencoders (SDAE) to achieve a remarkable 100% classification accuracy[12]. In another study, Acharya et al. (2018) implemented a 13-layer Convolutional Neural Network (CNN) on EEG recordings, reporting an accuracy of 88.67% while demonstrating the potential of deep learning models in classifying normal, pre-seizure, and seizure states[13]. On the other hand, Ömer Türk et al. (2019) achieved up to 100% accuracy in binary classifications by employing CNNs based on 2D frequency-time scalograms[14]. Comparative analyses by J. Wu et al. (2020) showcased the enhanced performance of CEEMD-XGBoost over previous models, highlighting significant improvements in sensitivity, specificity, and accuracy on both the Bonn and CHB-MIT datasets[15]. Moreover, Yang Liu et al. (2020) tested multiple dimensionality reduction techniques alongside deep learning, resulting in better recall, precision, and F1 scores than traditional methods, addressing challenges associated with data imbalance[16]. Recent research by Gurwinder Singh et al. (2020) emphasized the importance of feature relevance by utilizing multi-scale entropies and complete ensemble empirical mode decomposition (EEMD) in their hybrid approach, which enhanced classification

performance[17]. Additionally, Ibrahim Aliyu et al. (2021) demonstrated a significant reduction in the number of trainable parameters for LSTM networks through discrete wavelet transform (DWT) for noise removal and feature extraction[18]. Afshin Shoeibi et al. (2021) explored tunable-Q wavelet transform (TQWT) for sub-band extraction and implemented autoencoders for dimensionality reduction, achieving effective classification with ANFIS[19]. Al-Salman et al. (2022) extracted a set of signal features using the DT CWT (Dual-Tree Complex Wavelet Transform) and FFT methods. These features were then used to classify EEG signals from the Bonn and Bern-Barcelona datasets with the LS-SVM (Least Square Support Vector Machine) classifier[20]. In contrast, Fatima Hassan et al. (2022) combines CNN with machine learning classifiers, using feature selection methods to enhance accuracy, particularly on the Bonn and CHB-MIT datasets[21]. However, Wang et al. (2023) proposed a deep learning approach combining a convolutional neural network (CNN) and long short-term memory (LSTM) network for multi-class classification of epilepsy EEGs, including both binary and ternary tasks. The hybrid CNN-LSTM model automatically learns features without requiring manual preprocessing, benefiting from enhanced feature extraction and memory capabilities.[22]. Varli et al. (2023) takes a hybrid approach by transforming EEG signals into images using CWT and STFT, achieving high accuracy rates across multiple datasets, especially when combining time-sequence and time-frequency analyses[23]. Lastly, Sayani Mallick et al. (2024) proposed a novel method combining 1-D Convolutional layers with Bidirectional LSTM and Gated Recurrent Unit (GRU), achieving outstanding accuracy ranging from 95.81% to 100% across various classifications [24].

This work proposes a powerful deep-learning framework incorporating spatial and temporal multiscale stages. The first stage utilizes 1D convolutional layers combined with a continuous wavelet transform (CWT). This is followed by a 2D CNN, and the process finishes with a 1D Global Average Pooling layer.

The second stage includes a combination of GRU and double-layered LSTM, to capture both short-term and long-term temporal dependencies within the EEG data. This is essential for accurately classifying seizure events, as EEG signals contain patterns that span multiple time frames. In the next, followed by dense layers. Dropout layers separate the stages to prevent overfitting. The extracted features are then passed through dense layers, leveraging the strengths of CNNs and RNNs to enhance feature extraction and temporal sequence classification.

In this work, the sections are organized as follows: Section 2 describes the dataset and the experimental methodology. Section 3 presents the experimental results, while Section 4 offers a detailed discussion of the obtained findings. Finally, Section 5 concludes the paper and addresses the identified limitations.

II. METHODS AND MATERIALS

A. Dataset

EEG signal data from Bonn University in Germany were used in this study [25]. This dataset has been frequently utilized to identify epileptic seizures. The 500 EEG recording segments that make up the dataset are evenly split into five

sets A, B, C, D, and E. The characteristics of each cluster are given in Table I.

TABLE I. CLUSTERS AND PROPERTIES OF ELECTROENCEPHALOGRAM DATASETS USED IN THE STUDY.

Set	A	B	C	D	E
Patients	Healthy	Healthy	Epilepsy Patient	Epilepsy Patient	Epilepsy Patient
Total number of segments	100	100	100	100	100
Duration of each segment	23.6s	23.6s	23.6s	23.6s	23.6s
Description	Eyes open recording	Eyes closed recording	Pre-seizure record from the hippocampal half sphere	Pre-seizure record from the epileptic area	Record during the seizure

Electroencephalogram recordings were taken using the 10–20 international electrode positioning system. Each cluster consists of 100 parts with a single channel of 23.6 s.

The EEG signals used were filtered through a 0.53–40 Hz bandpass filter. The sampling rate of the filtered EEG signals is 173.61 Hz. The sample signals for these five clusters are shown in Figure 1. Then, the proposed cases are presented in Table II.

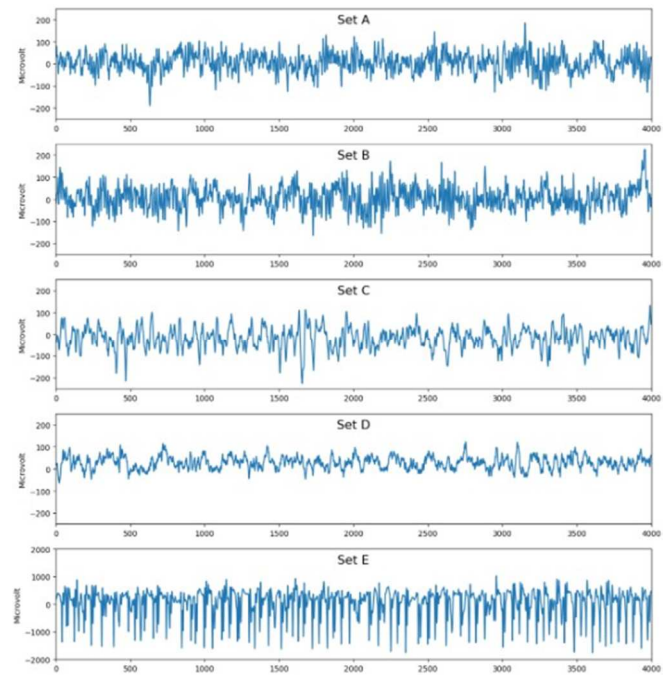


Fig. 1. EEG data sets used in this Study.

The proposed classification work is shown in Table II.

TABLE II. MORE DETAILS ABOUT THE FOUR PROBLEM CLASSIFICATION

Dataset	Classification	Description
Dataset1	CD, E	Interictal - Seizure
Dataset2	ABCD-E	Normal - Seizure
Dataset 3	AB, CD, E	Healthy - Interictal - Seizure
Dataset 4	A- D-E	Healthy - Interictal - Seizure

In this study, no pre-processing was applied to the data sets.

B. Deep Learning Models

1) Feature extraction in the spatial domain

A CNN architecture typically consists of an input layer, which receives raw time-series data EEG. The input layer guarantees that the data is properly formatted and normalized for subsequent processing stages.

a) 1D Convolutional Layer

After the input layer, we use a convolutional layer that is one dimension (Conv1D). The Conv1D layer applies a series of learnable filters across the temporal dimension to extract local features from the input data. The parameters of this layer, such as the number of filters and the kernel size (5), are chosen to optimize the feature extraction process. Each filter performs a convolution operation, sliding over the input data to capture local patterns and short-term dependencies. This layer outputs a series of feature maps that highlight significant local characteristics of the input signal according to Equation (1).

$$a_i = f(\sum_{m=1}^M w_m x_{i+m} + b) \quad (1)$$

Where a_i is the activation at position i , f is the non-linear function, w_m is the weight of the 1Dconvolution kernel, x_{i+m} is the input at position $i + m$, and b is the bias.

b) Continuous Wavelet Transform (CWT) layer

The output of the CWT layer is a time-frequency representation, which usually produces a 2D array with each entry representing a distinct time and frequency component of the input signal. As though The CWT transformation was applied to the EEG data set using the Morlet Continuous Wave. Since, Every segment yielded a scalogram image, so the analysis comprised 500 images in all (100 for A, 100 for B, 100 for C, 100 for D, and 100 for E). Examples of sample scalogram images for each cluster are shown in Figure 2.

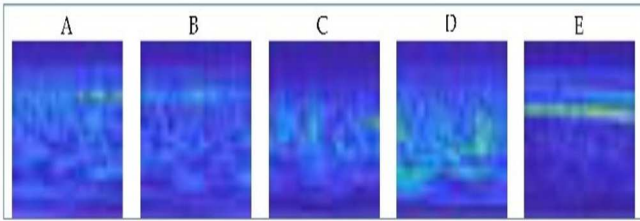


Fig. 2. scalogram images of EEGdata

c) 2D Max-Pooling Layer

This layer performs down sampling on the 2D representations from the CWT layer, to minimize dimensionality while preserving the most important features given in Equation (2).

$$a_{i,j}^{pooled} = \max_{m,n} (a_{i+m,j+n}) \quad (2)$$

Where $a_{i,j}^{pooled}$ is the pooled activation at position (i, j) and $\max$ is the max-pooling Operation over a window specified by m and n .

d) 2D Convolutional Layer

This layer extracts additional spatial information using Equation (3) and applies convolution operations to the 2D representations that are produced following max-pooling.

$$a_{i,j} = f(\sum_{m=1}^M \sum_{n=1}^N w_{m,n} x_{i+m,j+n} + b) \quad (3)$$

where $a_{i,j}$ is the respective activation, f is the non-linear activation function, $w_{m,n}$ is the weight matrix of convolution kernel $m \times n$, $x_{i+m,j+n}$ is the upper neuron activation connected to the neuron (i, j) , and b is the bias term. The pooling layer streamlines processes and reduces network parameters.

Following this, the 2D convolutional layer is connected to a max-pooling layer [26], batch normalization [27], and a Rectified Linear Unit function (ReLU) activation function [28]. The same structure is applied to the third convolutional layer (Conv2). The network concludes with a flattening layer [29], pooling layer [30], and a dropout layer to prevent overfitting [31].

2) Classification in the temporel domain

A Recurrent Neural Network (RNN) involves sequential processing of the data for learning. The ability of this sequential process to remember what came before the current sequence being processed justifies it. The reason it is called recurrent is that each time step's output is used as input for the subsequent time step. This in turn allows us to learn long-term dependencies in the training data.

RNN is composed of layers with memory cells. RNNs can use a variety of memory cell types. The Long Short-Term Memory (LSTM) unit or cell is one example of this sort [32]. When the sequence is processed at each time state, LSTM comprises a carry, a cell state. Ensuring that there is no loss of information during the sequential process is the responsibility of the carry. Figure 3 depicts the anatomy of an LSTM cell. An LSTM cell is composed of weights and three different gates for learning. At each time step, there is an input gate for the current input, an output gate to predict values and a forget gate that is used to discard information that is irrelevant.

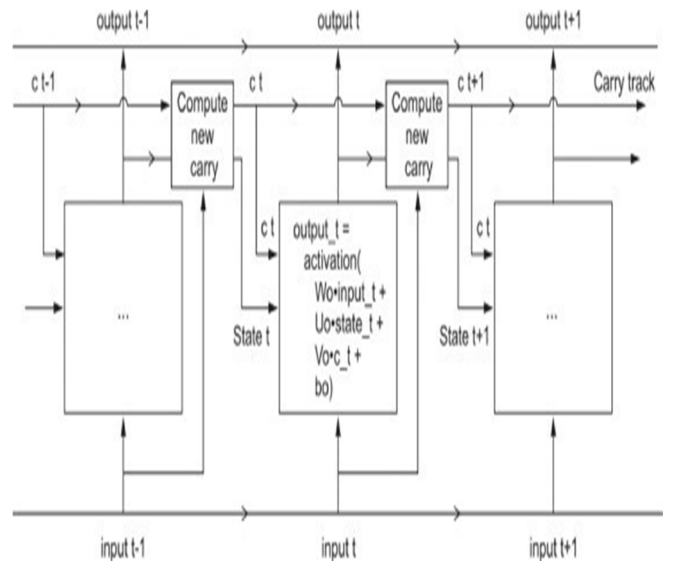


Fig. 3. The anatomy of an LSTM cell.

a) Gated recurrent units

Gated recurrent units are a form of recurrent neural network similar to LSTMs. They have the additional advantage of fewer trainable parameters [33]. The fact that they have fewer trainable parameters is an added benefit [33]. A GRU consists of a reset gate, an update gate, and a candidate activation function. The reset gate controls how much of the previous hidden state is forgotten, while the update gate controls how much of the new input is incorporated into the current hidden state. The new hidden state is then given to the following time step by the candidate activation function, which computes it using the input and the previous hidden state.

b) Hybrid CNN-RNN model

The proposed model leverages CNN's ability to extract local features and LSTM's capacity to learn long-term dependencies. The processing pipeline is as follows:

- A 1D convolutional (Conv1D) layer is first utilized to process input vectors and extract local features.
- Next, a Continuous Wavelet Transform (CWT) layer is employed to convert the output into a 2D map.
- The feature maps produced by the CNN layer serve as input for a Gated Recurrent Unit (GRU) layer, followed by two LSTM units/cells.
- The RNN layer uses the local features extracted by CNN to determine the long-term dependencies of these features.
- Finally, the model classifies the input data.

The architecture of the proposed model is illustrated in Figures. 4 and 5.

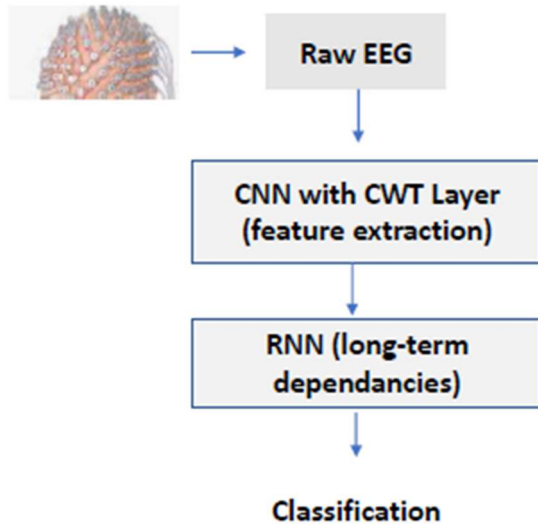


Fig. 4. The proposed hybrid model.

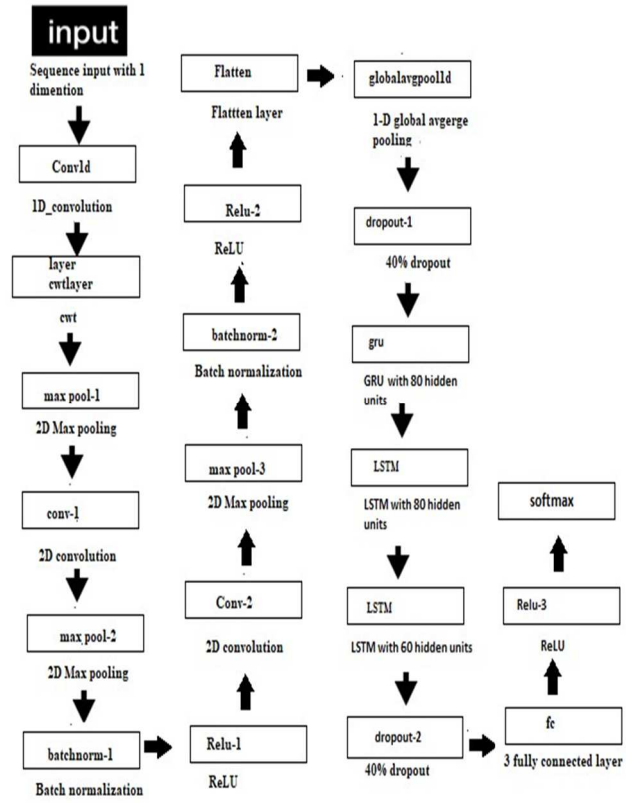


Fig.5. Block diagram of the proposed CNN-RNN network.

C. Performance Evaluation

Classification performance is calculated to determine how robust the suggested system. To compare the classification results in this study, we operated on the confusion matrix using: Accuracy formulated in Equation (4).

$$Acc = \frac{TP+TN}{TP+FP+FN+TN} \quad (4)$$

F1-Score is presented in Equation (5).

$$F - score = \frac{2*TP}{(2*TP + FP + FN)} \quad (5)$$

Sensitivity (SE) is formulated in Equation (6).

$$SE = \frac{TP}{TP+FN} \quad (6)$$

Specificity (SP) is formulated in Equation (7)

$$SP = \frac{TN}{FP+TN} \quad (7)$$

Precision is formulated in Equation (8)

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

III. RESULTS

The experiments in this study were performed on a PC equipped with a six-core Intel i7 processor using MATLAB (2023b). The server was equipped with an NVIDIA GEFORCE 920M with 6 GB of memory. This section presents results of the obtained were separated into 70% of the data set used for training, 20% of the data used for Testing, and 10% for validation.

The models were trained for 100 epochs, with the loss function optimized using the Adam optimizer. The learning rate was set to 0.001, and the batch size was one sample per iteration. Table III presents the number of images used in each stage of the training process.

TABLE III. NUMBER OF EEG SEGMENTS USED IN TRAINING, VALIDATION, AND TESTING IN BINARY AND THIRTY CLASSIFICATION

Data Set Taken into Consideration	Data set 1 CD, E	Data set 2 ABCD, E	Data set 3 AB, CD, E	Data set 4 A, D, E
Total Number	300	500	500	300
Number of segments Used for Training	210	350	350	210
Number of segments Used for Validation	30	50	50	30
Number of segments Used for Testing	60	100	100	60
Number of Classes at Proposed Model Output	2	2	3	3

TABLE IV. SUMMARIZE THE PERFORMANCE OF SUB-DATASETS OF BONN USED IN THIS STUDY

Dataset 1			Dataset 2			Dataset 3			Dataset 4		
	C	E		AB	E		A	C		A	D
D	D		CD			B	B	D			
C	39	1	AB	80	0	A	40	0	0	A	20
D			CD			B				D	0
E	0	20	E	0	20	D	1	39	0	E	0
						E	0	1	19		20

Table IV found different accuracy rates in the classification (validation and Test). while, In the validation phase of datasets 1, 2, 3, and 4, the accuracy of each dataset helps to adjust the model parameters and select the best model configuration by providing insight into the model performance on invisible data that was not used during training. This ensures that the model does not overfit the training data and can be generalized to new data. In the testing phase of each dataset, accuracy serves as the final performance indicator, evaluating the model's predictive power on a completely independent dataset.

Table V provides a comparative analysis of the predicted classes across four distinct datasets from Bonn. Additionally, by analyzing the results, one may determine which model consistently outperforms the others in terms of prediction efficacy and accuracy across a variety of datasets. This comparison is crucial for evaluating the generalizability and

robustness of the predictive models in handling diverse data characteristics inherent in the Bonn datasets.

TABLE V. COMPARISON OF PREDICT CLASSES OF FOUR DATASETS OF BONN

		Model	Validation Accuracy	Test Accuracy
Dataset1	CD, E	CNN_RNN	100%	98.33%
Dataset2	ABCD-E	CNN_RNN	100%	100%
Dataset 3	AB-CD-E	CNN_RNN	98.00%	98.00%
Dataset4	A- D-E	CNN_RNN	100%	100%

TABLE VI. COMPARISON PERFORMANCE OF ACCURACY, SENSITIVITY, PRECISION, AND F1- SCORE BETWEEN VARIOUS TEST APPROACHES IN A) DATASET 1,B) DATASET 2, DATASET 3, AND DATASET 4.

Dataset t		F1-score	Sensitivity	Specificity	Precision	Accuracy
Dataset t1	CD, E	98.14	98.75	98.75	97.61	98.33
Dataset t2	ABC D-E	100	100	100	100	100
Dataset t3	AB-CD-E	97.90	97.50	98.88	98.35	98.00
Dataset t4	A- D-E	100	100	100	100	100

The results of the classifier algorithms for each dataset are provided below. To assess the performance of the classification algorithms, each dataset was processed under identical conditions, and the evaluation parameters were reported as averages. Tables IV and V show the average classification results for different cases (datasets) of the Bonn dataset. the results indicate that in binary classification, the highest performance was achieved with dataset 2, reaching a 100% success rate. for the classification of healthy signals versus pre-seizure signals and seizure signals, dataset 4 performed better than dataset 3. This performance superiority of dataset 4 is confirmed by various parameters, including f1-score, sensitivity, specificity, and precision.

IV. DISCUSSION

Remarkably successful results are obtained with CNN_RNN architecture which can today be applied to many fields.

In the scientific world, it is predicted that deep learning networks, which add new methods every day, will achieve the desired success in many areas in the future. Additionally,

robust inference through deep learning is expected to improve the reliability of clinical decision support systems.

So far, various AI methods have been proposed for epileptic seizure detection, aiming to aid specialist physicians in the rapid diagnosis of epileptic seizures based on EEG signals. Researchers in the past have mostly exploited ML methods to diagnose epileptic seizures. Inefficiency in large amounts of input data, the complexity of processes, the need for great knowledge to use ML methods in diagnosing epileptic seizures, etc. are the most important deficiencies of these methods. To address this issue, in recent years, DL approaches have been proposed that possess appropriate efficiency and performance for diagnosing various diseases, including epileptic seizures, by using a large amount of input data.

The proposed method included two parts: feature extraction based on 2d CNN and classification by RNN network. First, the Conv1D layer captures local patterns in the data, while the CWT layer provides a multi-resolution analysis, enabling the model to extract both temporal and frequency-based features. It is conducted with a two-dimensional deep neural network consisting of a series of convolutional layers, a batch normalization layer, and a max-pooling layer, finished by 1d averg pooling layer. Next, The integration of GRU followed by double LSTM layers allows the model to handle both short-term dependencies (with GRU) and more complex long-term dependencies (with LSTMs). This hierarchical structure enables the extraction of diverse and rich features, from local signal patterns to global temporal relationships. The CWT introduces a new frequency-time dimension, enhancing the model's ability to deal with non-stationary signals, while the GRU-LSTM combination optimizes computational efficiency without sacrificing performance. Finally, the processed features are passed through fully connected layers with ReLU activations, and the softmax layer outputs the probability distribution for classification states.

This architecture stands out for its innovative integration of convolutional, wavelet, and recurrent networks, which work together in a highly complementary manner. This synergy significantly enhances performance and makes the model exceptionally well-suited for handling complex signals like EEG, enabling it to effectively capture both temporal dynamics and frequency-based features for more accurate analysis. Among the advantages of the proposed architecture, its high accuracy and higher efficiency in practical applications can be significant. The results showed that the proposed CNN-RNN scheme achieved the maximum level of accuracy among all the algorithms used.

Then, in Table VII, researchers conducted on Bonn datasets for epileptic seizure diagnosis using AI methods are presented and compared with the proposed method.

TABLE VII. THE REPORTED WORKS ON THE CLASSIFICATION OF BONN EEG SIGNAL.

Study	Method used	Data sets	Success %
[14]	CNN + Scalogram	Two_classes	99.50
		Multi_classes	99.00
[15]	Segmentation, CEEMD + Different Features + XGBoost	Two_classes	100
		Multi_classes	99.00
[16]	Spectral Thresholding +	Two_classes	---

	Different Features + RF	Multi_classes	98.80
[17]	TQWT + Fuzzy Entropy Features Set + AE + ANFIS-BS	Two_classes	99.74
		Multi_classes	99.46
[18]	CEEMD + MDE, RCMDE + Filter-Wrapper Based Method, One-Way-ANOVA + ANN	Two_classes	99.00
		Multi_classes	98.97
[19]	Convolutional RNN (CNN-RNN)(Statistical + Frequency + Fractal and Entropy Features)	Two_classes	- -
		Multi_classes	99.71
[20]	DT CWT (Dual-Tree Complex Wavelet Transform) and FFT methods+ LS-SVM (Least Square Support Vector Machine) classifier	Two_classes	- - -
		Multi_classes	96.8%
[21]	1d CNN+machine learning	Two_classes	100
		Multi_classes	94-99
[22]	convolution neural network (CNN) with a long short-term memory (LSTM) network	Two_classes	- -
		Multi_classes	98%
[23]	combined deep learning model by (CWT with scalograms and STFT with spectrograms) as input for the CNN (Convolutional Neural Networks) component, while the numerical signal data was used as input for the LSTM (Long Short-Term Memory) component.	Two_classes	99.07-99.28
		Multi_classes	97.06-98.56
[24]	1-D Convolutional layers in combination with Bidirectional Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) and Average pooling Layer as a single unit.	Two_classes	99 – 100
		Multi_classes	97.2 – 99.2
In this study	CNN-RNN	CD-E	98.33
		ABCD-E	100
		AB-CD-E	98
		A-D-E	100

Table VI shows that the proposed method for epileptic seizure diagnosis using 2d CNN features in the spatial domain and the proposed RNN model in the temporal domain has achieved good results. However, the proposed method outperforms other research projects. Table VII shows that the results are reliable and that it is possible to use this proposed method in clinical applications to diagnose epileptic seizures. The method proposed in this article is very effective in the

diagnosis of epileptic seizures. In this method, different handcrafted features are combined with DL, which improves the accuracy of epileptic seizure diagnosis based on EEG signals. The proposed method can help specialists quickly diagnose epileptic seizures. This study shows that the proposed method can be implemented on a software platform and used in hospitals.

V. CONCLUSION

Our study proposed a deep learning-based approach for detecting epileptic seizures using EEG recordings. Four datasets from Bonn University were used for training and testing, achieving high accuracies: 100% for datasets 2 and 4, 98.33% for dataset 1, and 98% for dataset 3.

There are two main aspects to this study. The first involves using EEG signals to identify seizures and pre-seizure (pre-ictal) or normal and seizure states. The second addresses the varying characteristics of different EEG datasets, which typically require the use of different features to achieve good performance across different classes. This traditional approach is not practical for handling EEG signal classification.

Our study differs from previous research in several ways. No dimensionality reduction techniques were applied, and the evaluation focused on binary and triple classification to assess the proposed method's success. The method successfully distinguished each dataset based on its unique characteristics. Our approach demonstrated significantly better performance compared to existing methods in the literature, especially for datasets 2 and 4. The results indicate that our method is highly effective in differentiating EEG signals.

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